01 / PROJECT INTRO

533 MILLION RELATIONS. NO EDGE TABLE.

A signed algebraic genealogy instead of a materialized graph.

533,208,060

K2048 qualified two-term ZD assessor edges

O(log N)

signed basis-oracle steps

Vladimír Bíža · Paper A + B · Rust / Python / one-click HTML

02 / THE PROBLEM

Graphs usually store answers.

What if relationships can be calculated directly from addresses?

Stored graph: edge, index, lookup. Large structures can be expensive to store, enumerate and evolve.

ZD approach: signed XOR recurrence + a four-address predicate + genealogy. General matrix-free operators also compress; the ZD atlas additionally specifies exact native algebraic structure.

03 / EXACT ALGEBRA

Seven colors.
Five kinds of query.

A K16 root generates inherited structures in higher dimensions.

01	ZERO / NONZERO
02	COMMUTATOR
03	ASSOCIATOR
04	ALTERNATOR
05	BIRTH / ANCESTRY

04 / THE HIDDEN CHANNEL

The pairwise product is zero. The triple interaction is not.

Algebraic information that boolean adjacency cannot encode.

$$x = e_2 + e_{11}$$

$$y = e_4 - e_{13}$$

$$xy = 0$$

$$[x, e_1, y] = -4e_7$$

Not cryptography. An exact higher-order algebraic channel.

05 / THE GENEALOGY

Inheritance is not enough. New edges are born.

Every dimension doubling carries old relations and qualified new ones.

4 × E(N)

inherited zero-product edges

≈ 50 %

asymptotic newborn fraction

$$E(N) = (N-2)(N-4)(N-8)/16$$

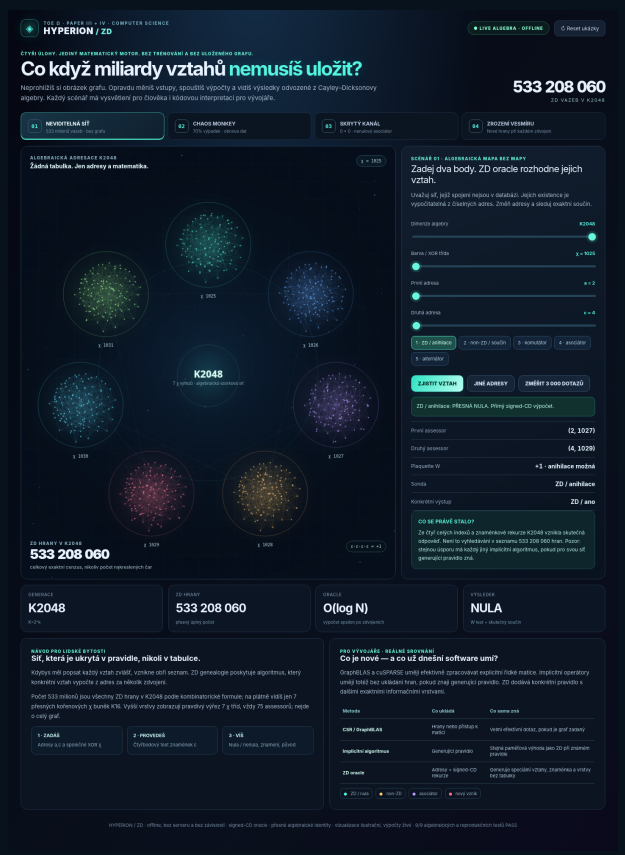
$$B(2N) = E(2N) - 4E(N)$$

This is not merely replication of the same table: new sectors require their own generative classification.

06 / TRY IT

Run the calculation.
Not just the animation.

Offline HTML + reproducible Python/Rust + two mathematics and computing papers.



GitHub: fractal-zd-computing

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